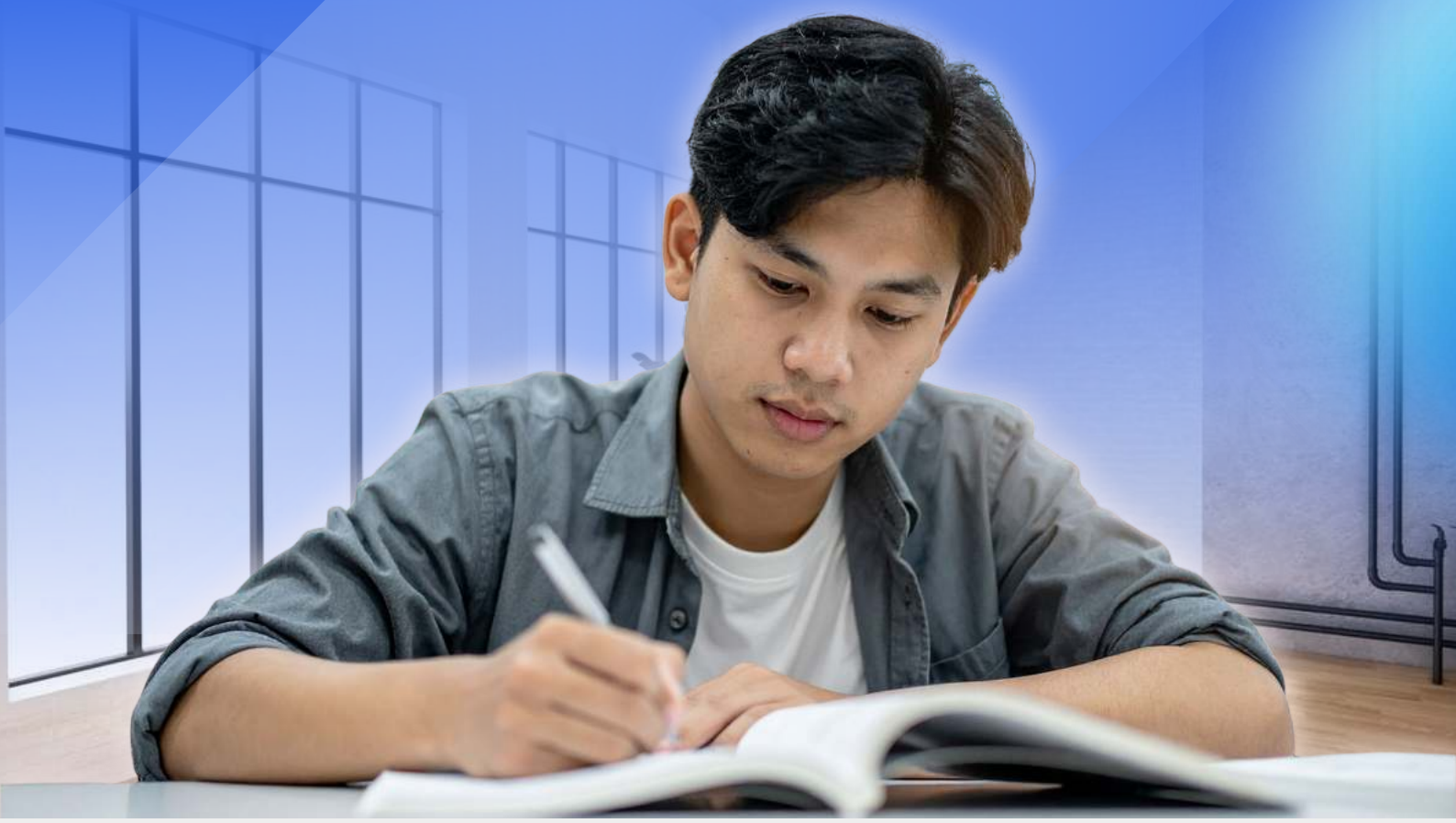
Lenterago

TRY OUT #1

SAT MATH

61002

44 QUESTIONS



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MODUL 1

22 QUESTIONS

DIRECTIONS

The questions in this section address a number of important math skills.
Use of a calculator is permitted for all questions.

NOTES

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- All figures lie in a plane.
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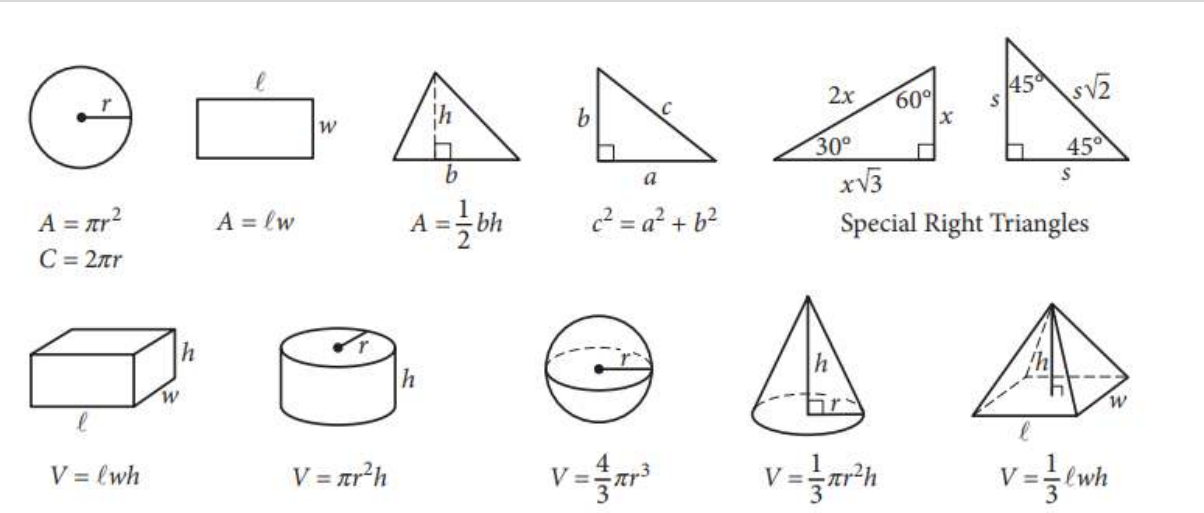

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2p.

The sum of the measures in degrees of the angles of a triangle is 180.

0 8 2 1 7 0 2 7 6 0 7 6

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Module 1

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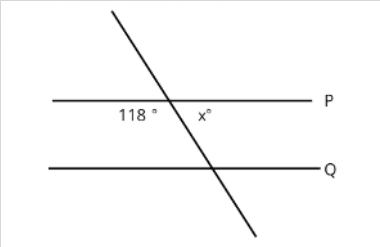
1. If $5y + 10 = 35$, which equation will have the same solution for y?
   A) $5y = 20$
   B) $5y = 15$
   C) $5y = 30$
   D) $5y = 10$

2. 

   In the figure below, line $p$is parallel to line q. If the measure of the angle shown is $118°$,what is the value of x?
   A) 62
   B) 30
   C) 40
   D) 160

3. The numbers below represent the ages (in years) of a group of children:
   5,8,10,12,15,20,8,5,7,10
   What is the mean age of the children?

4. The equation $2b + 5r = 100$describes the relationship between the number of birds $b$and the number of reptiles $r$that can be cared for at a pet shop in one day. If the shop cares for 10 reptiles, how many birds can it care for?
   A) 10
   B) 15
   C) 20
   D) 25

5. A cube has a side length of 20 inches. What is the volume of the cube in cubic inches?
   A) 400
   B) 800
   C) 8.000
   D) 20.000
   Correct answer: C

6. 12 is q%of 40. What is the value of q?

7. Solve the equation:
   $$-2x^2 - 5x = -12$$
   What is the positive solution for x?
   A) $\frac{3}{2}$
   B) 2
   C) 3
   D) 4

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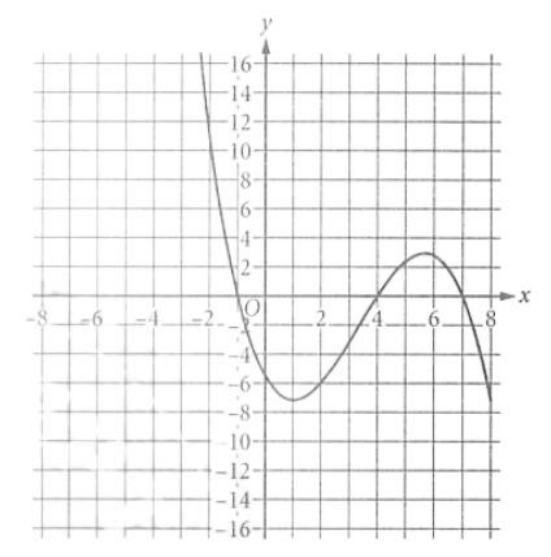
8. In 2010, the population of Riverdale was 20,000. A model predicts that each year, for the next 6 years, the population $p$grows by 5% compared to the previous year. Which of the following equations best represents this situation, where $x$is the number of years after 2010 and $x \leq 6$?
   A) $p = 20{,}000(0.95)^x$
   B) $p = 20{,}000(1.05)^x$
   C) $p = 1.05(20{,}000)^x$
   D) $p = 0.95(20{,}000)^x$

9. Two classes, Class X and Class Y, each recorded the number of pens brought by students.
   In Class X, the numbers ranged only from 3 to 7 pens.
   In Class Y, the numbers ranged from 12 to 20 pens.
   Which statement best compares the standard deviations of the two classes?
   A) The standard deviation for Class X is less than the standard deviation for Class Y.
   B) The standard deviation for Class X is equal to the standard deviation for Class Y.
   C) The standard deviation for Class X is greater than the standard deviation for Class Y.
   D) There is not enough information to compare.

10.



The graph of $y = f(x)$is shown, where the function is

$$f(x) = ax^3 + bx^2 + cx + d,$$

and $a, b, c, d$are constants. For how many values of $x$does $f(x) = 0$?
Options:
A) One
B) Two
C) Three
D) Four

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11. The exponential function is defined by

$$h(x) = 25 \cdot b^x,$$

where $b$is a positive constant. If $h(2) = 400$,what is the value of $h(3)$?

12. Solve the system of equations:

$$y = (x-1)(x+2)y = 2x$$

Which ordered pair $(x, y)$is the solution?

A) $(-2, -4)$
B) $(0, 0)$
C) $(2, 4)$
D) $(3, 6)$

13. Triangles FGH and JKL are similar. Angle F corresponds to angle J. Angles G and K are right angles. If $\sin(F) = \frac{308}{317}$, what is $\sin(J)$?

A) $\frac{75}{317}$
B) $\frac{308}{317}$
C) $\frac{317}{308}$
D) $\frac{317}{75}$

14. A bacteria count is modeled by

$$f(t) = 60{,}000 \cdot 2^{t/410},$$

where t is time in minutes. How many minutes does it take for the population to double?

15. The positive numbers p, k, jsatisfy

$$p = \frac{k}{4j+9}.$$

Which equation correctly expresses $4j + 9$in terms of $p$and k?

A) $4j + 9 = \frac{k}{p}$
B) $4j + 9 = kp$
C) $4j + 9 = k - p$
D) $4j + 9 = \frac{p}{k}$

16. Line $p$is given by the equation $8x + 4y = 6$. Line $r$is perpendicular to line $p$in the coordinate plane. What is the slope of line r?

17. Point O is the center of a circle. The measure of arc RS on this circle is 100° .What is the measure, in degrees, of the central angle ROS associated with this arc?

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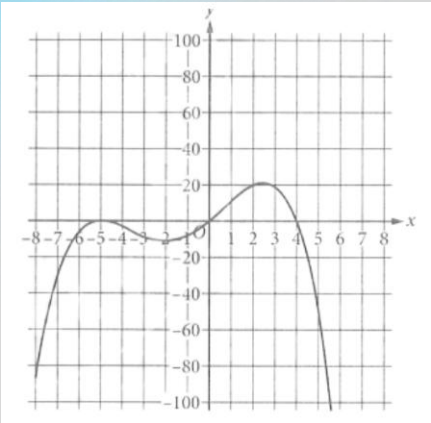
18. 

Which of the following equations could represent the graph shown in the coordinate plane?

A) $y = -\frac{1}{10}x(x-4)(x+5)$

B) $y = -\frac{1}{10}x(x-4)\left(x+5\right)^2$

C) $y = -\frac{1}{10}x(x-5)(x+4)$

D) $y = -\frac{1}{10}x\left(x-5\right)^2(x+4)$

19. For $x > 0$, the function f is defined as:

$$f(x) = 0.2x$$

Which of the following best describes this function?

A) Decreasing exponential
B) Decreasing linear
C) Increasing exponential
D) Increasing linear

20. Given

$$f(x) = 4x^2 + 64x + 262,$$

define $g(x) = f(x+5)$ .For what value of $x$does $g(x)$ reach its minimum?

A) -13
B) -8
C) -5
D) -3

21. A poll of 803 voters shows:
Angel Cruz: 483 votes
Terry Smith: 320 votes
If 6,424 voters participate in the real election, how many votes would Angel Cruz be expected to receive?

A) 163
B) 1,304
C) 3,864
D) 5,621

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22. Solve the system:

$$y = 2x^2 - 21x + 64y = 3x + a$$

where $a$is a constant. The graphs intersect at exactly one point. What is the value of $x$at that point?

A) -8
B) -6
C) 6
D) 8

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MODUL 2

22 QUESTIONS

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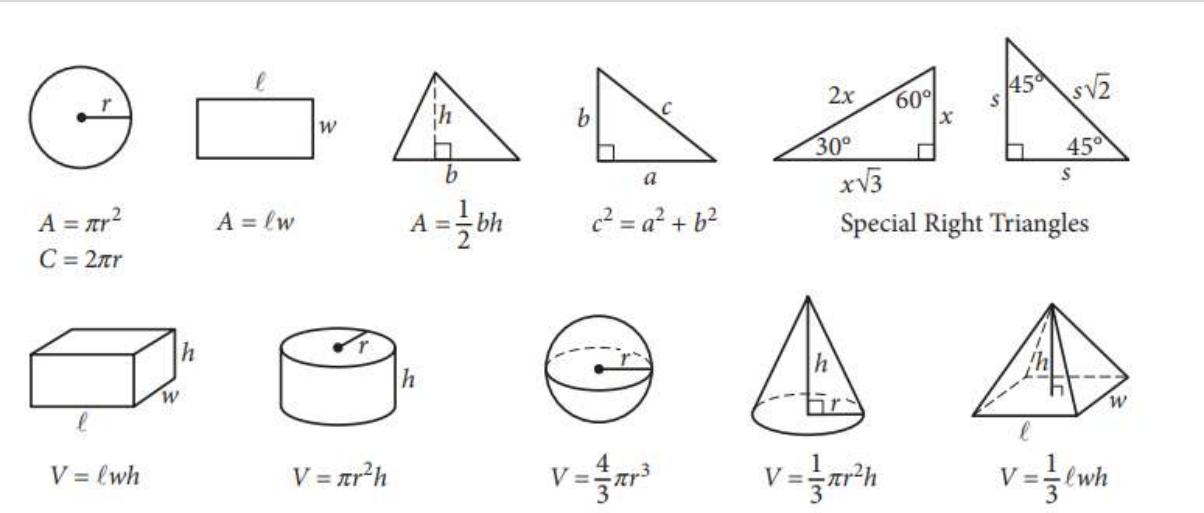

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23. Solve the system below:

$$2x = 10 - 2x + y = -4$$

What is the value of y?
A) -2
B) 6
C) 10
D) 14

24. Which of the following is equivalent to

$$9x^4 - 7x^4?$$

A) $16x^4$
B) $2x^4$
C) $2x^8$
D) $16x^8$

25. How many solutions does the equation

$$45y = 45y$$

have?
A) One
B) Two
C) Infinitely many
D) None

26. In a city vote about building a new library, a report said the number of "yes" votes was three times the number of "no" votes. A social media post also claimed there were 15,000 more "yes" votes than "no" votes. Based on these statements, how many people voted no?
A) 7,500
B) 15,000
C) 22,500
D) 45,000

27. Caleb makes popsicles using fruit juice. The amount of juice left (in fluid ounces) after making $x$popsicles is modeled by

$$f(x) = -5x + 30.$$

Which statement best describes the y-intercept of this graph in this context?
A) Caleb used about 5 fluid ounces of juice per popsicle.
B) Caleb had about 5 fluid ounces of juice at the start.
C) Caleb had about 30 fluid ounces of juice at the start.
D) Caleb used about 30 fluid ounces of juice per popsicle.

28. An angle measures $\frac{16\pi}{15}$radians. What is its measure in degrees?

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29. Solve the system of inequalities:

$$y \leq 2x + 5,\ y \geq -x - 3$$

Which point is a solution?

A) $(-5, 0)$

B) $(0, -5)$

C) $(0, 6)$

D) $(5, 0)$

30. A right triangle has legs of 9 cm and 12 cm. If its hypotenuse can be written in the form $3\sqrt{d}$ cm (where $d$is an integer), what is the value of d?

31. Two data sets each have 20 values. The possible values are $10,20,30,40,50$.
In Set A, higher numbers occur more often.
In Set B, lower numbers occur more often.
Which statement is correct about the means of the two sets?

A) The mean of Set A is greater than the mean of Set B.

B) The mean of Set A is less than the mean of Set B.

C) The means are equal.

D) Not enough information.

32. Triangle $ABC$is similar to triangle DEF, with $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$. If $\angle C = 30^\circ$ and $DE = 2 \cdot AB$, what is the measure of $\angle F$?

A) 10°

B) 20°

C) 30°

D) 40°

33. Maria prepared 40 cups of lemonade. She filled $x$small bottles and $y$large bottles with all the lemonade. The equation

$$2x + 5y = 40$$

represents this situation. What does the term $5y$represent in this context?

A) The number of large bottles Maria filled

B) The number of small bottles Maria filled

C) The total number of cups of lemonade in the large bottles

D) The total number of cups of lemonade in the small bottles

34. Solve the equation:

$$x(x + 3) = 10$$

What is the sum of the solutions?

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35. The function:

$$h(x) = \frac{1}{8}(x-6)^2 + 4$$

represents the height (in feet) of a ball at time $x$seconds after it is thrown, where $0 \leq x \leq 12$. Which of the following best describes the vertex of this parabola?
A) The ball's minimum height was 4 feet above the ground.
B) The ball's minimum height was 6 feet above the ground.
C) The ball's height was 4 feet above the ground when it started moving.
D) The ball's height was 6 feet above the ground when it started moving.

36. The function

$$T(k) = \frac{9}{5}(k-273) + 32$$

converts a temperature of $k$kelvins to degrees Fahrenheit. If the temperature increases by 5kelvins, how many degrees Fahrenheit does the temperature increase?
A) 5
B) 7
C) 9
D) 15

37. A line passes through the points $(m, 20)$ and $((m+4), (-4))$ .The line crosses the y-axis at $(0, c)$ .What is the value of c?
A) 12
B) 20
C) 28
D) 36

38. The polynomial

$$x^3 + 9x^2 + 20x$$

has a factor of the form $(x+b)$,where $b$is a positive integer. What is the smallest possible value of b?

39. The function $g(x) = 5x - 35$ .What is the x-intercept of the graph of $y = g(x)$in the coordinate plane?

A) $(-7, 0)$
B) $(0, -7)$
C) $(7, 0)$
D) $(0, 7)$

40. A rectangular field has an area of 6,195,200 square feet. What is this area in square miles?
($1\ mile = 5280\ feet$)
A) 0.22
B) 0.44
C) 0.88
D) 1.22

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41. One gallon of paint covers 300square feet. If a wall has a total area of $w$square feet and needs two coats, how many gallons of paint $P$are required?
A) $P = \frac{w}{150}$
B) $P = \frac{w}{300}$
C) $P = 150w$
D) $P = 300w$

42. Consider the system of equations:

$$36x - 54y = 18 + 24x$$

$$sy = 12x$$

where s is a constant. If this system has no solution, what is the value of s?

43. Solve for x:

$$\frac{x^2}{\sqrt{x^2 - d^2}} = \frac{d^2}{\sqrt{x^2 - d^2}} + 25$$

where $d > 0$. Which of the following is a solution?

A) $-d$

B) $-\sqrt{25^2 - d^2}$

C) $\sqrt{25^2 - d^2}$

D) $\pm\sqrt{25^2 + d^2}$

44. Let

$$g(x) = ax^2 + 6x + b,$$

where $a > 0$. The parabola $y = g(x)$opens upward and has vertex (h, k) .If $h < 0$and $g(-8) = g(x)$,which statements must be true?

I. $b < 0$
II. $a \geq 1$

A) I only
B) II only
C) I and II
D) Neither I nor II

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